Stat 415 Lab 5 - Explainability

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1 Introduction

This report will focus on using attribution methods to explain the outputs of our models. Attribution methods weight the relative importance of inputs with respect to making a prediction, and the ones we will be using in this report are LIME, Shapley Values, and SmoothGrad.

2 The Data

We perform analyses on the following two datasets:

1. *Pima Indians Diabetes Database*: This tabular database has several health diagnostic measures such as blood pressure, and the goal is to predict if a patient has diabetes.
2. *Animal-10N Image Classification Dataset:* The ANIMAL-10N dataset consists of ten classes of animals, comprising 50,000 training images and 5,000 testing images. Your task is to develop a model capable of accurately classifying the animal depicted in a given image.

2.1 Diabetes Dataset

The first dataset has 768 observations and 9 variables, which are shown in Table 1.

Variable	**Description**
Pregnancies	Number of times the patient has been pregnant
Glucose	Plasma glucose concentration measured 2 hours in an oral glucose tolerance test
BloodPressure	Diastolic blood pressure (mm Hg)
SkinThickness	Triceps skinfold thickness (mm), a measure of body fat
Insulin	2-Hour serum insulin (mu U/ml)
BMI	Body Mass Index (weight in kg / (height in m)^2)
DiabetesPedigreeFunction	A function that scores likelihood of diabetes based on family history
Age	Age of the patient (in years)
Outcome	Class variable (1 = diabetes, 0 = no diabetes)

Table 1: Description of variables in the Pima Indians Diabetes Dataset

We notice that among the 768 samples, 500 patients are non-diabetic and 268 are diabetic. In other words, 34.90% of the patients are diabetic. In a balanced dataset, we would expect to see close to a 50/50 split. However, here, we see that the dataset is moderately unbalanced since the diabetic class is underrepresented.

2.2 Image Classification Dataset

The Animal-10N dataset is a real-world image classification benchmark containing 55,000 images across 10 animal classes (e.g., cat, dog, elephant, etc.), collected from the internet with natural label noise introduced by human annotators. It is widely used for studying noisy label learning and robust image classification methods.

3 Interpreting Models that Predict Diabetes

3.1 Logistic Regression

To predict the diabetes binary outcome variable, we first fit a logistic regression model with an L1 penalty. Table 2 shows the selected features and their coefficients. Since all eight variables were selected even after regularization, we infer that each variable contributes some predictive value to the model.

We notice that glucose, BMI, and age have the strongest positive association with diabetes. Also, interestingly, higher blood pressure and higher levels of insulin have a negative correlation with diabetes, although the effect sizes are pretty small.

Feature	**Coefficient (L1 Penalty)**
Pregnancies	0.2088
Glucose	1.0599
BloodPressure	-0.2425
SkinThickness	0.0269
Insulin	-0.1826
BMI	0.7854
DiabetesPedigreeFunction	0.2183
Age	0.4208

Table 2: Selected features and their coefficients using logistic regression with L1 regularization

3.2 Random Forest

We also fit a random forest to predict the binary diabetes outcome variable. Then, we extracted the feature importance coefficients to understand the level of influence each feature has on the likelihood of getting diabetes.

We see similar results as we saw for logistic regression: glucose was the best predictor of diabetes, followed by BMI and then age. We also see that insulin and blood pressures are very insignificant predictors of diabetes. Lastly, we see that pregnancies is a significantly worse predictor when using the random forest than in the logistic regression model.

Feature	**Importance Score**
Pregnancies	0.0766
Glucose	0.2589
BloodPressure	0.0881
SkinThickness	0.0656
Insulin	0.0761
BMI	0.1700
DiabetesPedigreeFunction	0.1238
Age	0.1409

Table 3: Feature importances from Random Forest classifier

3.3 LIME (Local Interpretable Model-agnostic Explanations)

We use the LIME (Local Interpretable Model-agnostic Explanations) explainer to better understand the importance of each feature in our random forest model. Note that we focus on patient 0 from the diabetes dataset and use LIME without discretization (continuous variables are not discretized).

As seen in Table 4, we see that, like seen in the importance scores for the random forest model and the coefficients for the logistic regression model with an L1 penalty, one's glucose levels have the strongest correlation with their likelihood of having diabetes. This is followed by BMI, then age. We also see that blood pressure is the only variable with a negative contribution, which is consistent with the logistic regression model having a negative coefficient for this feature. Insulin also had a negative coefficient in the logistic regression model, but in our analysis using the LIME explainer, we see that insulin has a near zero contribution to diabetes, suggesting that rather than having a negative relationship, Insulin has closer to no relationship to the chance of getting diabetes. The other variables seem to moderately contribute to diabetes.

We also notice that the next five patients also have the same order of features in terms of their level of contribution to the outcome. This shows that the LIME explainer approximates the random forest model consistently across different data points.

Feature	**LIME Contribution (No Discretization)**
Pregnancies	0.0171
Glucose	0.1332
BloodPressure	-0.0059
SkinThickness	0.0103
Insulin	0.0018
BMI	0.0945
DiabetesPedigreeFunction	0.0522
Age	0.0661

Table 4: LIME explanation for test instance 0 (continuous features, no discretization)

3.4 SHAP (SHapley Additive exPlanations)

We then used SHAP (SHapley Additive exPlanations) to assess the infleunce of each of the features. We see that the results are consistent with LIME, random forest, and logistic regression. Glucose, BMI, and age in that order, are the best predictors of diabetes. Furthermore, we see that insulin and blood pressure are again, weak predictors of diabetes.

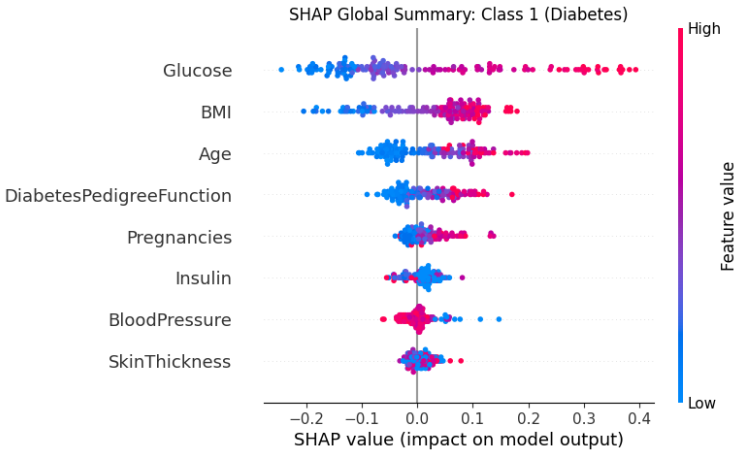


Figure 1: SHAP Global Feature Importance Across Whole Dataset

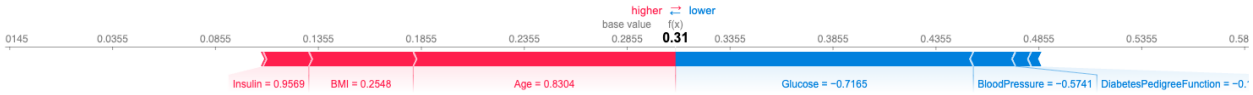


Figure 2: Force Plot for Patient 0

We see in Figure 2 the SHAP force plot for patient 0. This plot explains the model's prediction for an individual patient in terms of how each feature influenced the final probability of having diabetes (class 1). The base value of the model—the average predicted probability across all patients—is approximately 0.31. The model predicted a probability of about 0.31 for this patient as well, meaning the individual's risk is average. Features like Insulin (0.9569), BMI (0.2548), and Age (0.8304) pushed the prediction toward diabetes (shown in red), increasing the risk. Meanwhile, features like Glucose (-0.7165), BloodPressure (-0.5741), and DiabetesPedigreeFunction (-0.4634) pushed the prediction away from diabetes (shown in blue), lowering the risk. In this case, the positive and negative contributions nearly cancel each other out, leading to a final output close to the base probability. This plot demonstrates how SHAP provides a clear, individualized breakdown of feature influence on the model's decision.

3.5 Comparing Everything

The comparison table presents feature importances across four interpretability techniques—Random Forest Gini importance, Logistic Regression coefficients, SHAP values, and LIME values—using the Pima Indians Diabetes dataset. The Random Forest and Logistic Regression columns represent global feature importance, indicating which features generally have the most influence across all predictions. In contrast, SHAP and LIME provide local explanations for a single test instance, showing how much each feature contributed to that specific prediction. In the Random Forest and Logistic Regression columns, Glucose, BMI, and Age appear as the most influential features, consistent with clinical expectations. For the selected test instance, SHAP and LIME also attribute importance to BMI, Glucose, and Age, albeit with slightly different magnitudes and signs, reflecting the individualized nature of these methods.

The color gradient in the table visually emphasizes the relative strength of feature contributions within each method. Darker or more saturated shades represent higher absolute importance, making it easier to see which features each method prioritizes. For example, Glucose is clearly the darkest (and therefore most important) in both the Random Forest and Logistic Regression columns, while SHAP highlights Age and BMI more for the specific instance. This visual structure helps address the original question: "How do LIME, SHAP, Forest importance, and linear model weighting compare?" Overall, while all methods tend to agree on key features like Glucose and BMI, local methods like SHAP and LIME can diverge from global rankings, emphasizing that individual-level explanations may differ significantly depending on the specific input values. This underscores the complementary nature of global vs. local interpretability techniques in machine learning.

Table 5: Feature Importance Comparison Across Methods

Feature	RandomForest Importance	LogReg Coef (L1)	SHAP (local)	LIME (local)
Age	0.160411	0.420812	0.134519	0.066247
Glucose	0.342675	1.059890	-0.107169	0.115467
BMI	0.181538	0.785380	0.064504	0.078678
Insulin	0.068941	-0.182599	0.043814	0.015122
Pregnancies	0.060014	0.208765	0.012245	0.024899
BloodPressure	0.053263	-0.242467	-0.009443	0.000466
DiabetesPedigreeFunction	0.086599	0.218338	0.004219	0.036307
SkinThickness	0.046559	0.026896	0.000455	0.007143

4 Predictive Modeling on Animal Images

Our next objective is to do predictive modeling on animal images. Note that the training dataset is comprised of images like the one in Figure 3.

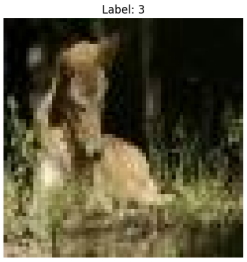


Figure 3: Training Example from Animal Classification Dataset

4.1 Linear Regression

We trained a single-layer linear classifier using PyTorch. We first processed raw image tensors from Animal10N's training split, and normalized them by dividing pixel values by 255.0 to scale into [0,1]. Then, we flattened the image shape from (3,43,43) to a flat vector of shape (5547,). After this data preprocessing, we trained the model (with 90% of the data used for training and 10% for validation) for 10 epochs with batch size 128, learning rate 0.001, Adam optimizer, and used the Cross Entropy loss.

After we trained the model, we validated it and get the training loss and validation accuracy per epoch as shown in Figure 4. The training loss seems to reduce, showing effective optimization. However, the validation accuracy is extremely low overall, fluctuating between 0.005 and 0.03. This reveals the limited predictive power of this model - it is underfitting.

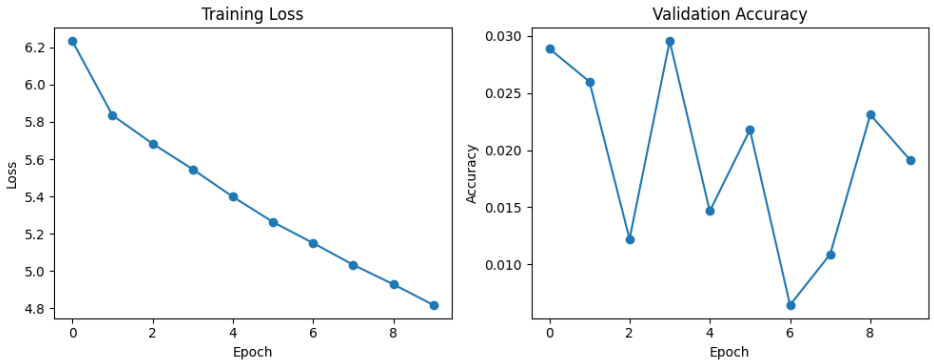


Figure 4: Training Loss and Validation Accuracy of Linear Classifier

4.2 Convolutional Neural Network

Next, we trained a vanilla Convolutional Neural network, with 3 convolutional blocks, Adam Optimizer with learning rate $1e-3$, cross entropy loss, batch size = 64, 15 epochs, and run on the CPU.

We also fine tuned this model by adding an extra convolutional block, added batch-normalization after every convolution, added dropout(0.5) in the FC block, and lowered the learning rate for the Adam optimizer to $5e-4$.

We can see in Figure 5 the difference in the training losses for the linear classifier, the vanilla CNN, and the fine-tuned CNN. We see that the Vanilla CNN learns the fastest and ist he most effective on the training data. The tuned CNN is slower to train and may require more epochs or hyperparameter adjustment or learning rate tuning to outperform the other two models. Lastly, the linear model does poorly overall, showcasing how it cannot be used for a task as complex as image classification.

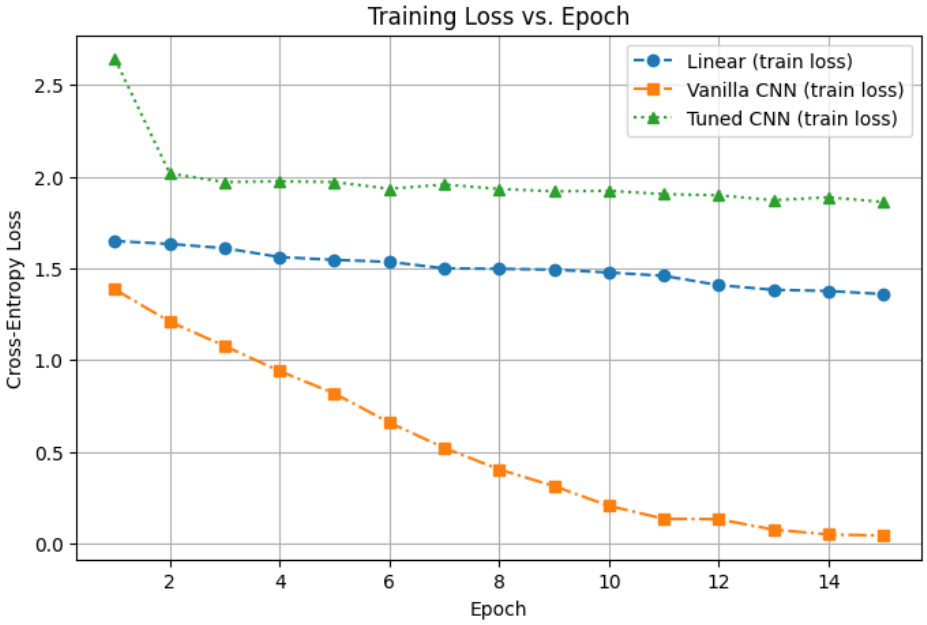


Figure 5: Training Loss vs Epoch for Linear Classifier, Vanilla CNN, and Tuned CNN

5 Feature Attribution on Animal Images

The last component of our analysis is feature attribution on animal images

5.1 SmoothGrad

We firstly ran SmoothGrad to our vanilla CNN, and get the heatmap, or SmoothGrad Saliency map, of the gradient on the images, shown in Figure 6. The left image is the raw input image, and the right shows which pixels contributed most to the model's prediction of the image (by computing the average of gradients over noisy perturbations). We see that the highlighted regions are well-distributed across the image, suggesting that the model is learning from multiple regions rather than overfitting to isolated features.

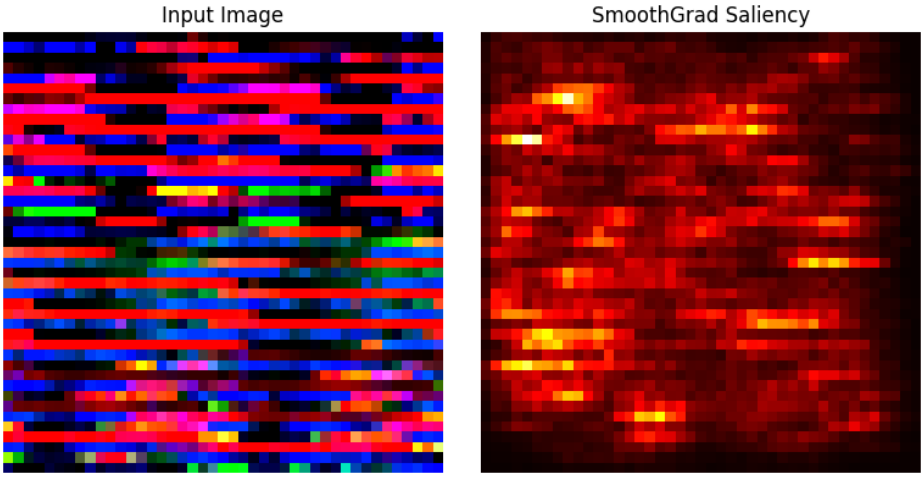


Figure 6: SmoothGrad Saliency Map

5.2 LIME

As seen in Figure 7, we can also use the LIME explainer to see which features most contributed to predicting the image. Note that the LIME map segments the image into interpretable superpixels and highlights those (in yellow) that most positively influenced the model's predicted class. LIME provides sparser, more localized insights than SmoothGrad, isolating specific structural patterns or textures the model relied on. In this case, LIME selected several central horizontal patterns and one vertical segment, possibly outlining fur texture or boundaries.

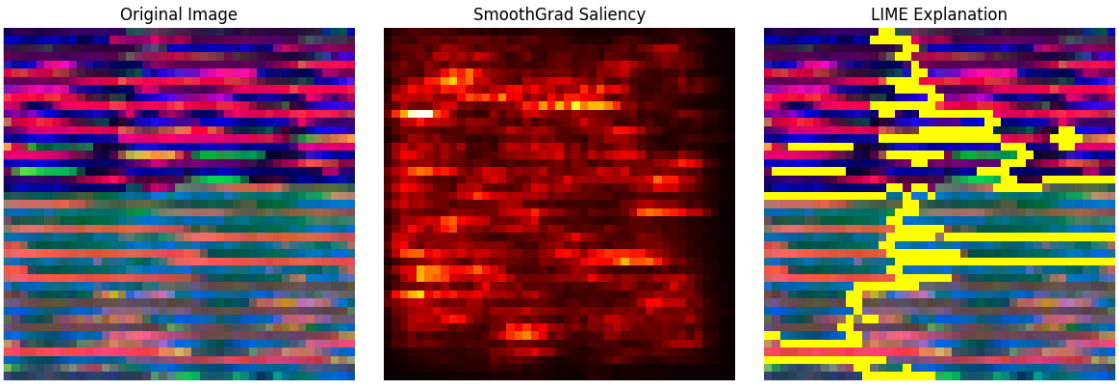


Figure 7: Comparing Raw image, SmoothGrad Saliency, and LIME explainer